Ejercicio 3.2: Imputación de datos perdidos (simple)

Silvia Pineda

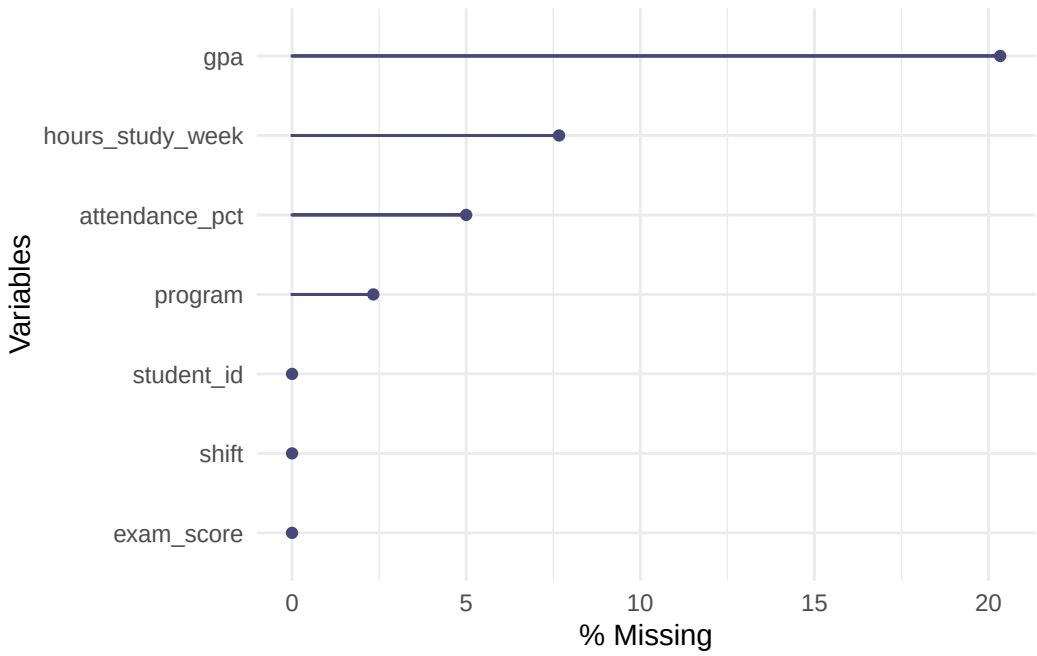
Carga de Datos y Librerías

```r
library(naniar)
library(tidyverse)
```

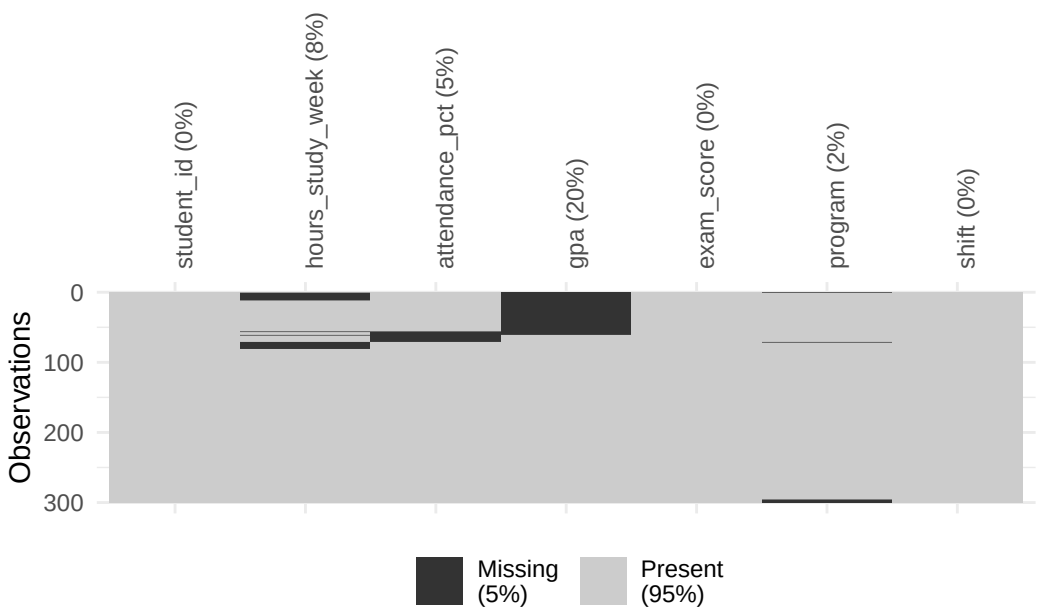
```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(rio)

data <- read.csv("students_FP.csv",
  na.strings = c("", "NA", "NaN", "NULL"),
  stringsAsFactors = TRUE
)
```

```r
gg_miss_var(data, show_pct = TRUE)
```




```
vis_miss(data,cluster = TRUE)  +
  theme(axis.text.x = element_text(angle = 90))
```



1.Imputación de las variables MCAR

La variable `program` tienen $< 5\%$ y es cualitativs, por tanto la forma más simple de imputarlas sería por la moda.

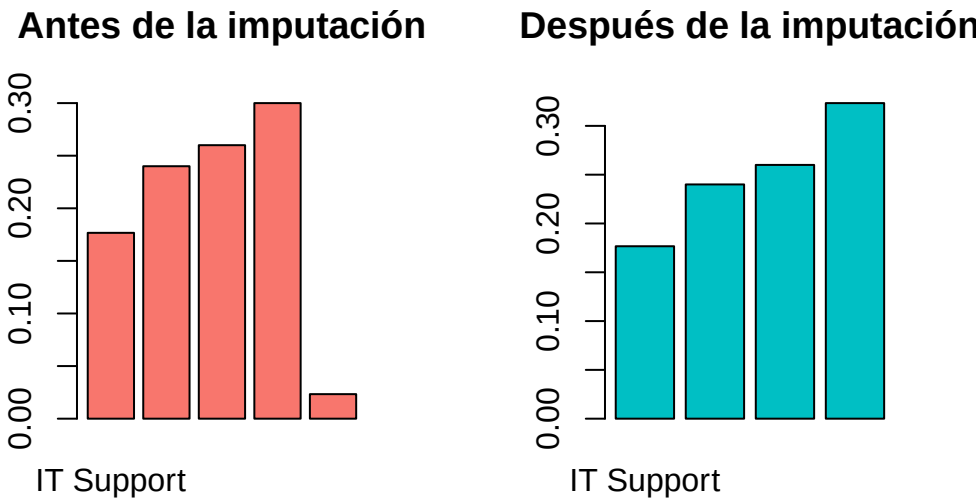
```
#Comprobar la categoría más frecuente
prop.table(table(data$program,useNA = "always"))
```

```
            IT Support Network Administration   Software Engineering
            0.17666667             0.24000000             0.26000000
       Web Development                   <NA>
            0.30000000             0.02333333
```

```
## Sustituir los NA con la categoría más frecuente
data$program_imp <-data$program
data$program_imp[is.na(data$program_imp)] <- "Web Development"

## Volvemos a comprobar
prop.table(table(data$program_imp,useNA = "always"))
```

```
            IT Support Network Administration   Software Engineering
             0.1766667              0.2400000              0.2600000
       Web Development                   <NA>
             0.3233333              0.0000000
```

```
# Graficar la distribución de la variable categórica antes y después de la imputación
par(mfrow = c(1, 2)) # Organizar las gráficas en una fila de 2 columnas
barplot(prop.table(table(data$program, useNA = "ifany")),
        main = "Antes de la imputación", col = "#F8766D" )
barplot(prop.table(table(data$program_imp)),
        main = "Después de la imputación",col = "#00BFC4")
```



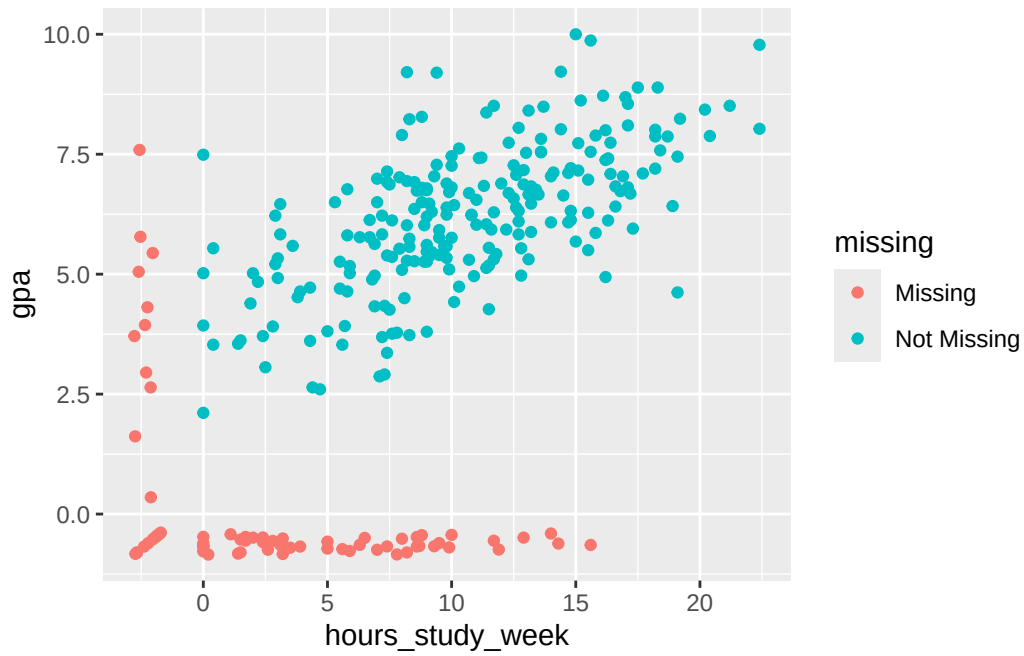
El % de datos missing de la variable program es muy bajo y al imputar por la moda, no cambia la distribución en absoluto, por tanto, lo asumiremos como una buena imputación

2.Imputación de las variables MAR/MNAR

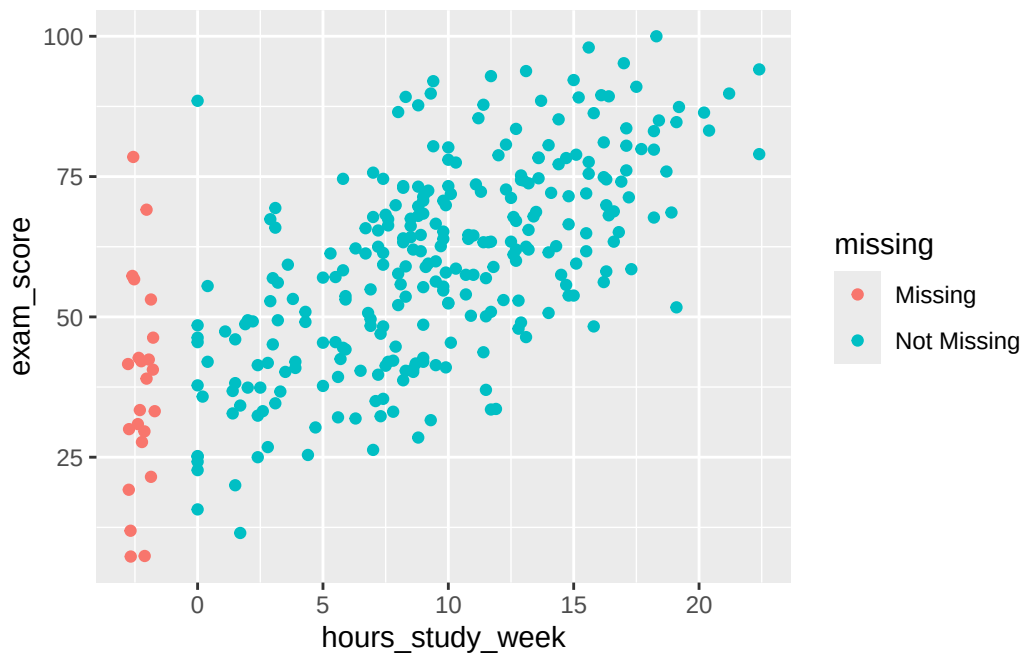
- Variable `hours_study_week`

Los NA de esta variable son MAR/MNAR y corresponden a un 8%, además la variable está asociada con gpa y exam_score:

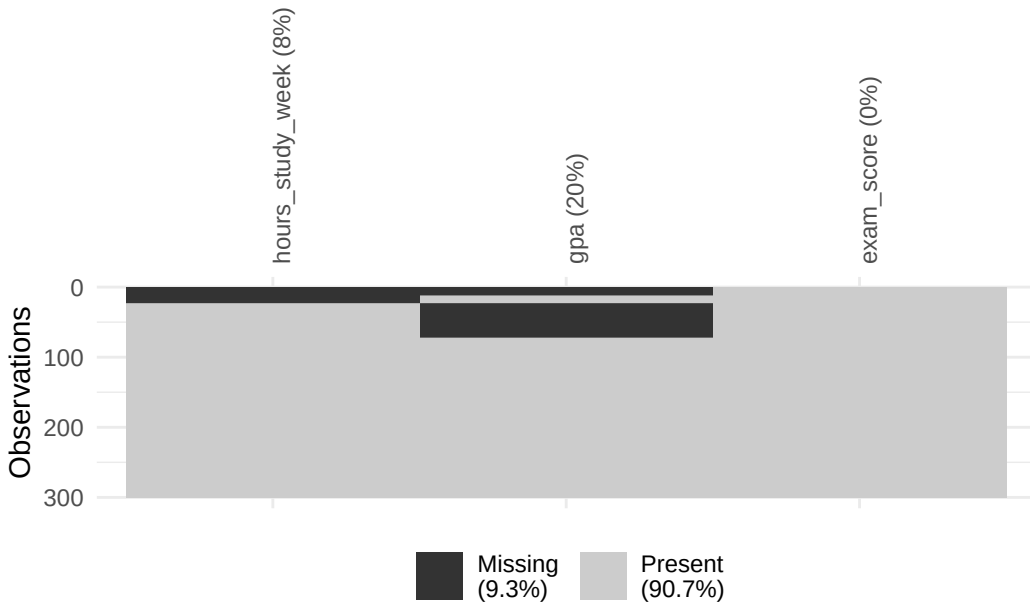
```
ggplot(data = data, aes (x = hours_study_week ,
                         y =gpa )) + geom_miss_point()
```




```
ggplot(data = data, aes (x = hours_study_week ,
                         y =exam_score )) + geom_miss_point()
```



```r
vis_miss(select(data,hours_study_week,gpa,exam_score),cluster=TRUE)  +
  theme(axis.text.x = element_text(angle = 90))
```



Como solo están completos los datos de `exam_score` para los NA de `hours_study_week`, la imputaremos mediante una regresión con esa variable:

```r
#Ajustar un modelo de regresión lineal
model1 <- lm(hours_study_week ~ exam_score, data = data)
summary(model1)
```

```
Call:
lm(formula = hours_study_week ~ exam_score, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-15.0151  -2.8472   0.0736   2.6104  10.9111 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.40111    0.82399   -1.70   0.0902 .  
exam_score   0.18549    0.01322   14.03   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

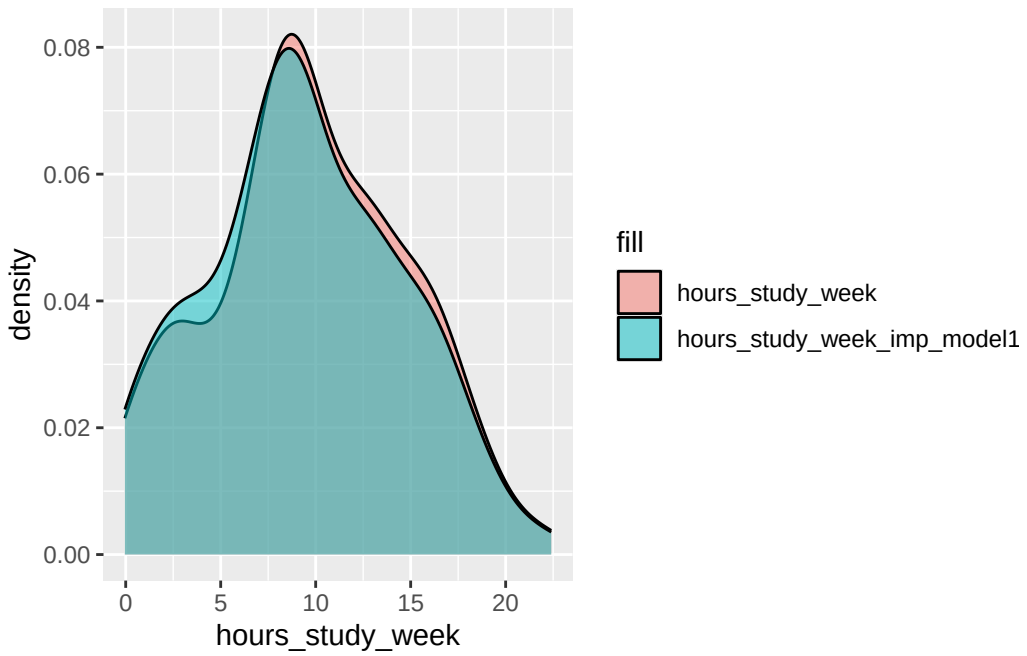
```
Residual standard error: 3.906 on 275 degrees of freedom
  (23 observations deleted due to missingness)
Multiple R-squared:  0.4172,    Adjusted R-squared:  0.4151
F-statistic: 196.9 on 1 and 275 DF,  p-value: < 2.2e-16
```

```r
#Predecir los valores solo para las observaciones faltantes
predictions <- predict(model1,newdata = data [is.na(data$hours_study_week),])

##Crear una nueva variable imputada
data$hours_study_week_imp_model1 <- data$hours_study_week
data$hours_study_week_imp_model1[is.na(data$hours_study_week)]<- predictions

summary(model1)
```

```
Call:
lm(formula = hours_study_week ~ exam_score, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-15.0151  -2.8472   0.0736   2.6104  10.9111 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.40111    0.82399   -1.70   0.0902 .  
exam_score   0.18549    0.01322   14.03   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.906 on 275 degrees of freedom
  (23 observations deleted due to missingness)
Multiple R-squared:  0.4172,    Adjusted R-squared:  0.4151
F-statistic: 196.9 on 1 and 275 DF,  p-value: < 2.2e-16
```

```r
#Hacer un gráfico para comparar las observaciones
ggplot(data, aes(x = hours_study_week, fill = "hours_study_week")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = hours_study_week_imp_model1, fill = "hours_study_week_imp_model1"), al
```

```
Warning: Removed 23 rows containing non-finite outside the scale range
(`stat_density()`).
```

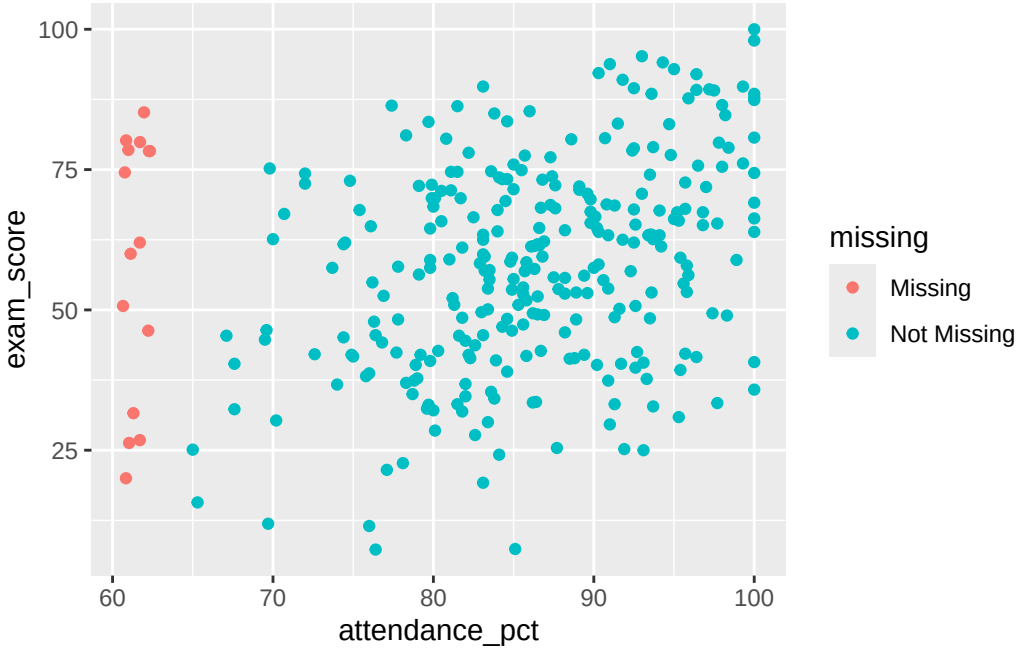


La variable se imputa bastante bien teniendo en cuenta que como faltan los datos de `hours_study_week` que presuponemos más bajos ya que están relacionados con datos más bajos de `exam_score`, la distribución tiene un leve desplazamiento a la izquierda.

- Variable `attendance_pct`

Los `NA` de `attendance_pct` son MNAR, pero con un % pequeño y además esta variable está asociada a exam_score y todos sus missing están completos

```
ggplot(data = data, aes (x = attendance_pct ,
                         y =exam_score )) + geom_miss_point()
```



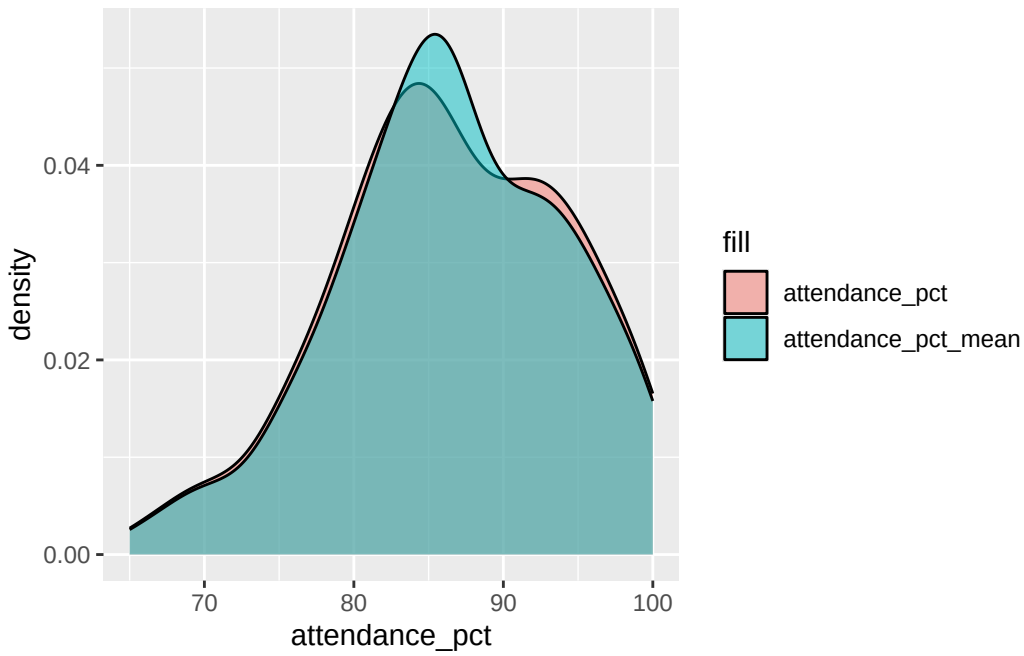
Como no tenemos muy claro que haya una asociación, vamos a realizar dos cosas, una probar la media y otra probar un modelo de regresión. Para el modelo de regresión como solo están completos los datos de `exam_score` para los NA de `attendance_pct`, probaremos con esa:

```
data$attendance_pct_mean<-data$attendance_pct #Nueva variable
mean(data$attendance_pct, na.rm = TRUE) #Media
```

```
[1] 86.45123
```

```
# 86.45123

data$attendance_pct_mean[is.na(data$attendance_pct_mean)] <- mean(data$attendance_pct_mean, r

##Gráfico para representar las diferencias
ggplot(data, aes(x = attendance_pct, fill = "attendance_pct")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = attendance_pct_mean, fill = "attendance_pct_mean"), alpha = 0.5)
```

```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_density()`).
```

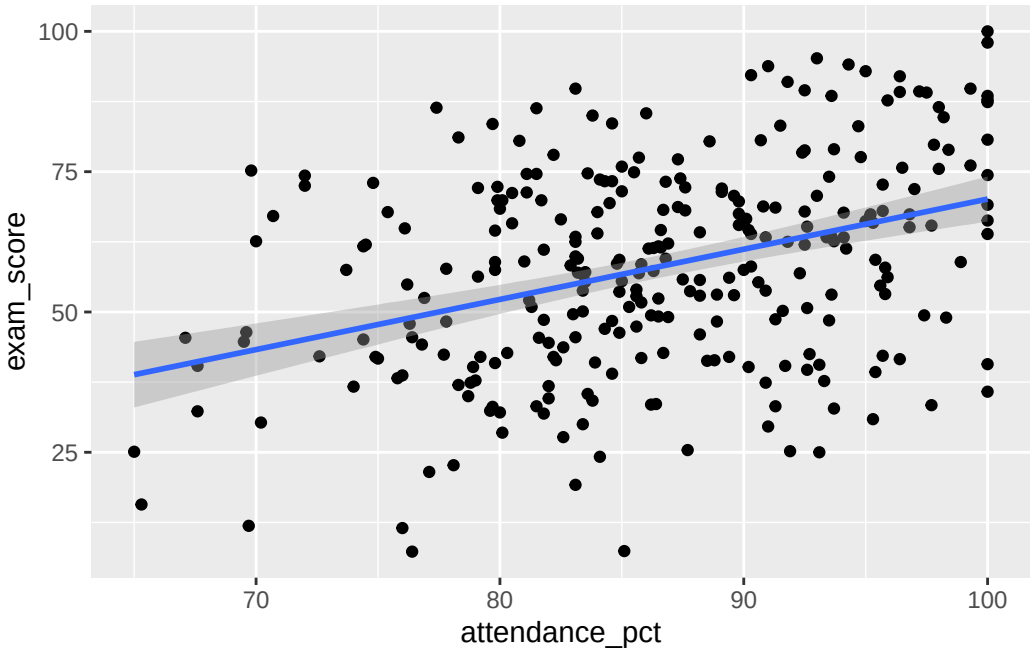



```r
#Ajustar un modelo de regresión lineal
model1 <- lm(attendance_pct ~ exam_score, data = data)
summary(model1)
```

```
Call:
lm(formula = attendance_pct ~ exam_score, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-19.4024  -4.3054  -0.0134   5.5110  17.1033 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 77.16726    1.41751  54.439  < 2e-16 ***
exam_score   0.16004    0.02328   6.874 3.98e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.269 on 283 degrees of freedom
  (15 observations deleted due to missingness)
Multiple R-squared:  0.1431,	Adjusted R-squared:  0.1401 
F-statistic: 47.26 on 1 and 283 DF,  p-value: 3.977e-11
```

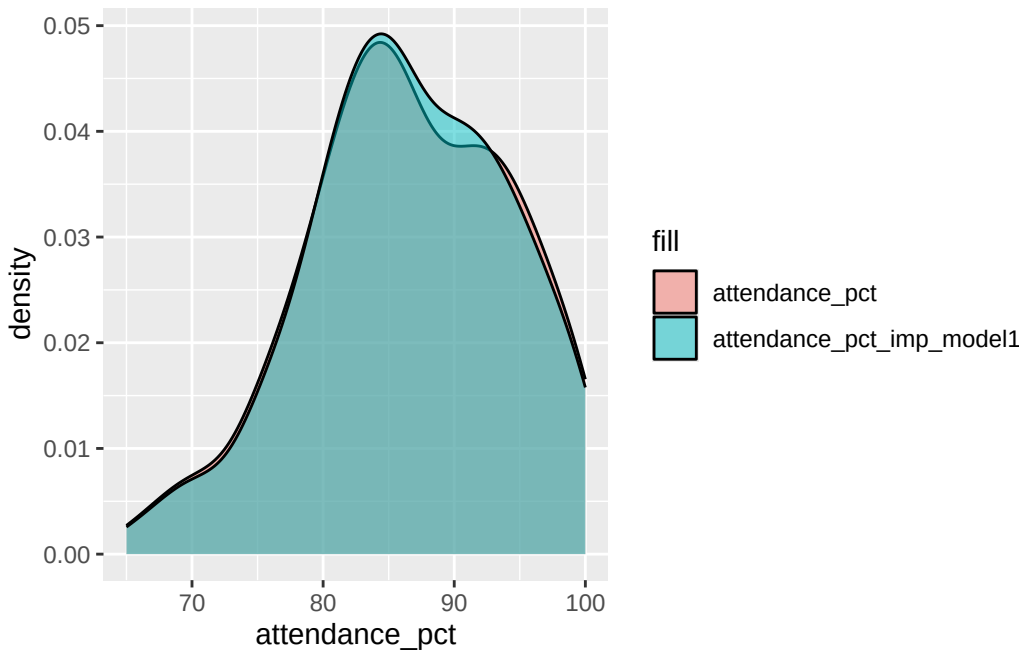
```r
ggp <- ggplot(data,aes(attendance_pct, exam_score)) + geom_point()
ggp + stat_smooth(method = "lm",
formula = y ~ x,
 geom = "smooth")
```

```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 15 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```r
#Predecir los valores solo para las observaciones faltantes
predictions <- predict(model1,newdata = data [is.na(data$attendance_pct),])

##Crear una nueva variable imputada
data$attendance_pct_imp_model1 <- data$attendance_pct
data$attendance_pct_imp_model1[is.na(data$attendance_pct_imp_model1)]<- predictions

#Hacer un gráfico para comparar las observaciones
ggplot(data, aes(x = attendance_pct, fill = "attendance_pct")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = attendance_pct_imp_model1,
                   fill = "attendance_pct_imp_model1"), alpha = 0.5)
```

```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_density()`).
```

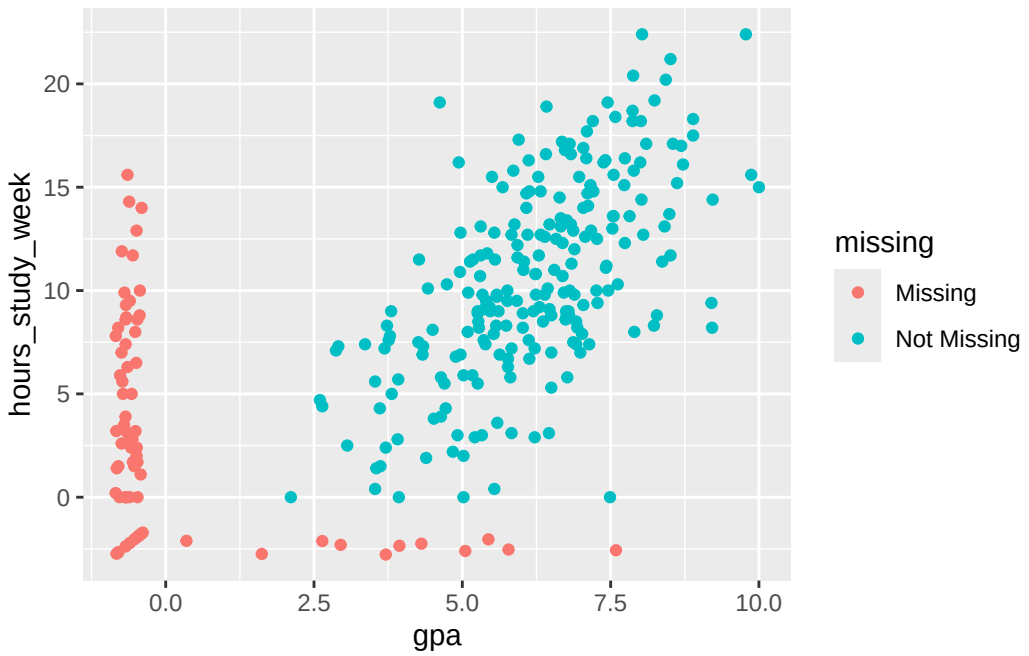


Como son pocos datos y hay una leve asociación, la imputación no parece demasiado incorrecta.

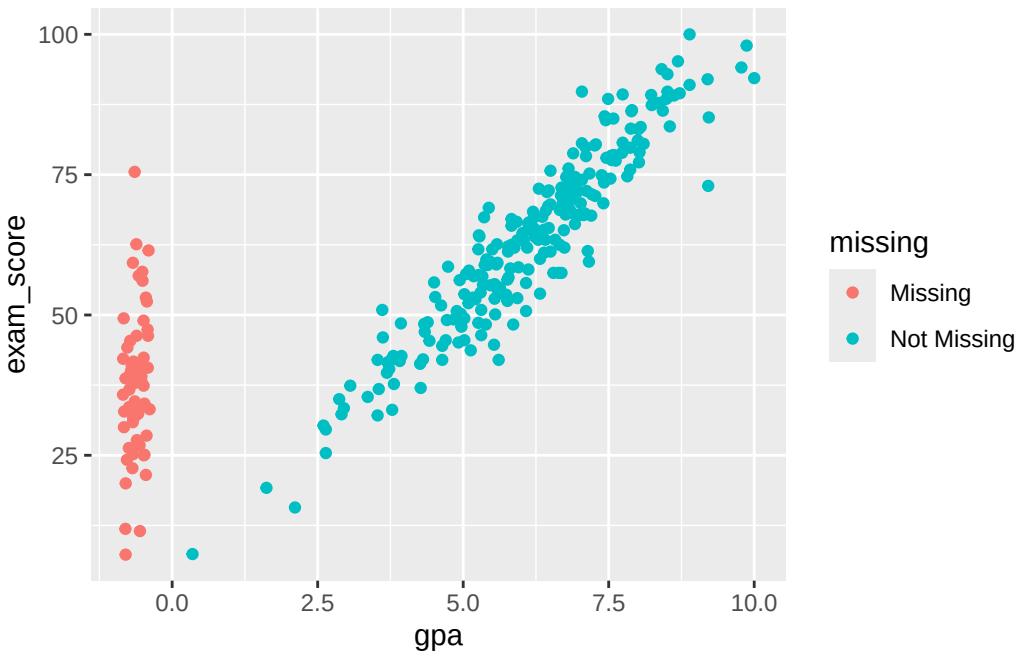
- Variable `gpa`

Los `NA` de `gpa` corresponden mayoritariamente a valores con datos inferiores de `hours_study_week` y con datos menores de `exam_score`.

```
ggplot(data = data, aes (x = gpa ,
                         y =hours_study_week )) + geom_miss_point()
```




```
ggplot(data = data, aes (x = gpa ,
                         y =exam_score )) + geom_miss_point()
```



En este caso, no tenemos ninguna de las dos variables asociadas como completas. Por tanto una opción es usar una de las ya imputadas. En este caso, usaremos exam_score que tiene el

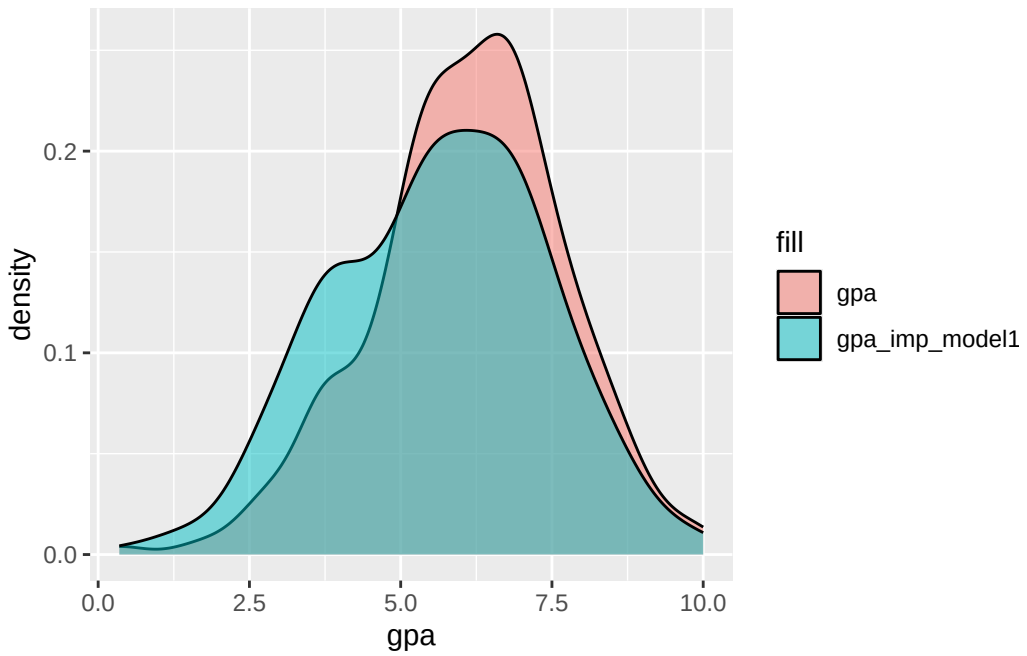
menor número de datos faltantes y una asociación muy clara

```r
#Ajustar un modelo de regresión lineal
model1 <- lm(gpa ~ exam_score, data = data)
summary(model1)
```

```
Call:
lm(formula = gpa ~ exam_score, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.49257 -0.33300 -0.02257  0.28643  2.20431 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.371042   0.135401    2.74  0.00661 ** 
exam_score  0.090886   0.002075   43.81  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5256 on 237 degrees of freedom
  (61 observations deleted due to missingness)
Multiple R-squared:  0.8901,	Adjusted R-squared:  0.8896 
F-statistic:  1919 on 1 and 237 DF,  p-value: < 2.2e-16
```

```r
#Predecir los valores solo para las observaciones faltantes
predictions <- predict(model1,newdata = data [is.na(data$gpa),])

##Crear una nueva variable imputada
data$gpa_imp_model1 <- data$gpa
data$gpa_imp_model1[is.na(data$gpa_imp_model1)]<- predictions

#Hacer un gráfico para comparar las observaciones
ggplot(data, aes(x = gpa, fill = "gpa")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = gpa_imp_model1, fill = "gpa_imp_model1"),
               alpha = 0.5)
```

```
Warning: Removed 61 rows containing non-finite outside the scale range
(`stat_density()`).
```



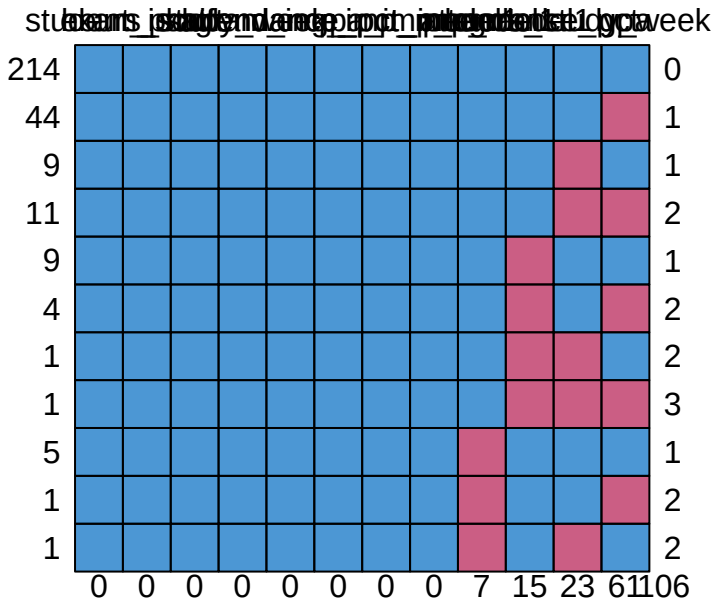
En este caso vemos que la imputación no es tan buena como en las anteriores, pero claro tenemos un 20% de datos y todos se imputan por los valores bajos de gpa.

3. Imputación múltiple con MICE

```
library(mice)
```

```
Attaching package: 'mice'

The following object is masked from 'package:stats':

    filter

The following objects are masked from 'package:base':

    cbind, rbind
```

```
md.pattern(data)
```




```
    student_id exam_score shift program_imp hours_study_week_imp_model1
214          1          1     1           1                           1
44           1          1     1           1                           1
9            1          1     1           1                           1
11           1          1     1           1                           1
9            1          1     1           1                           1
4            1          1     1           1                           1
1            1          1     1           1                           1
1            1          1     1           1                           1
5            1          1     1           1                           1
1            1          1     1           1                           1
1            1          1     1           1                           1
             0          0     0           0                           0
    attendance_pct_mean attendance_pct_imp_model1 gpa_imp_model1 program
214                   1                          1               1       1
44                    1                          1               1       1
9                     1                          1               1       1
11                    1                          1               1       1
9                     1                          1               1       1
4                     1                          1               1       1
1                     1                          1               1       1
1                     1                          1               1       1
5                     1                          1               1       0
1                     1                          1               1       0
1                     1                          1               1       0
```

```
                     0                          0                0        7
    attendance_pct hours_study_week gpa
214              1                 1   1   0
44               1                 1   0   1
9                1                 0   1   1
11               1                 0   0   2
9                0                 1   1   1
4                0                 1   0   2
1                0                 0   1   2
1                0                 0   0   3
5                1                 1   1   1
1                1                 1   0   2
1                1                 0   1   2
                15                23  61 106
```

No hay ninguna observación con todas las variables missing.

```
impData <- mice(select(data,hours_study_week,attendance_pct,gpa,exam_score,program, shift),m=
```

```
 iter imp variable
  1   1  hours_study_week  attendance_pct  gpa  program
  1   2  hours_study_week  attendance_pct  gpa  program
  1   3  hours_study_week  attendance_pct  gpa  program
  1   4  hours_study_week  attendance_pct  gpa  program
  1   5  hours_study_week  attendance_pct  gpa  program
  2   1  hours_study_week  attendance_pct  gpa  program
  2   2  hours_study_week  attendance_pct  gpa  program
  2   3  hours_study_week  attendance_pct  gpa  program
  2   4  hours_study_week  attendance_pct  gpa  program
  2   5  hours_study_week  attendance_pct  gpa  program
  3   1  hours_study_week  attendance_pct  gpa  program
  3   2  hours_study_week  attendance_pct  gpa  program
  3   3  hours_study_week  attendance_pct  gpa  program
  3   4  hours_study_week  attendance_pct  gpa  program
  3   5  hours_study_week  attendance_pct  gpa  program
  4   1  hours_study_week  attendance_pct  gpa  program
  4   2  hours_study_week  attendance_pct  gpa  program
  4   3  hours_study_week  attendance_pct  gpa  program
  4   4  hours_study_week  attendance_pct  gpa  program
  4   5  hours_study_week  attendance_pct  gpa  program
  5   1  hours_study_week  attendance_pct  gpa  program
```

```
  5   2  hours_study_week  attendance_pct  gpa  program
  5   3  hours_study_week  attendance_pct  gpa  program
  5   4  hours_study_week  attendance_pct  gpa  program
  5   5  hours_study_week  attendance_pct  gpa  program
  6   1  hours_study_week  attendance_pct  gpa  program
  6   2  hours_study_week  attendance_pct  gpa  program
  6   3  hours_study_week  attendance_pct  gpa  program
  6   4  hours_study_week  attendance_pct  gpa  program
  6   5  hours_study_week  attendance_pct  gpa  program
  7   1  hours_study_week  attendance_pct  gpa  program
  7   2  hours_study_week  attendance_pct  gpa  program
  7   3  hours_study_week  attendance_pct  gpa  program
  7   4  hours_study_week  attendance_pct  gpa  program
  7   5  hours_study_week  attendance_pct  gpa  program
  8   1  hours_study_week  attendance_pct  gpa  program
  8   2  hours_study_week  attendance_pct  gpa  program
  8   3  hours_study_week  attendance_pct  gpa  program
  8   4  hours_study_week  attendance_pct  gpa  program
  8   5  hours_study_week  attendance_pct  gpa  program
  9   1  hours_study_week  attendance_pct  gpa  program
  9   2  hours_study_week  attendance_pct  gpa  program
  9   3  hours_study_week  attendance_pct  gpa  program
  9   4  hours_study_week  attendance_pct  gpa  program
  9   5  hours_study_week  attendance_pct  gpa  program
  10   1  hours_study_week  attendance_pct  gpa  program
  10   2  hours_study_week  attendance_pct  gpa  program
  10   3  hours_study_week  attendance_pct  gpa  program
  10   4  hours_study_week  attendance_pct  gpa  program
  10   5  hours_study_week  attendance_pct  gpa  program
```

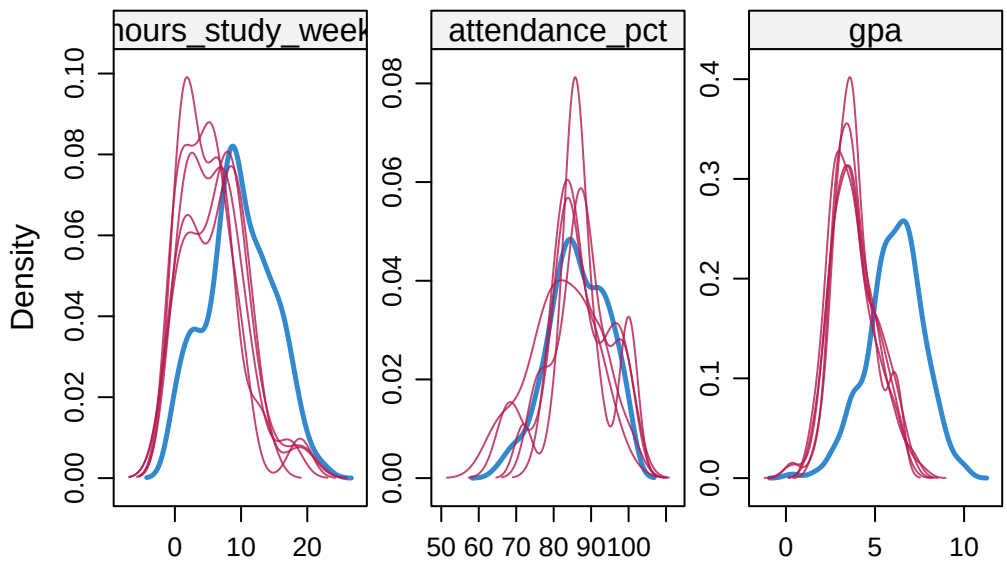
```r
summary(impData)
```

```
Class: mids
Number of multiple imputations:  5
Imputation methods:
hours_study_week   attendance_pct              gpa       exam_score
           "pmm"            "pmm"            "pmm"               ""
         program            shift
       "polyreg"               ""
PredictorMatrix:
                 hours_study_week attendance_pct gpa exam_score program shift
hours_study_week                0              1   1          1       1     1
```

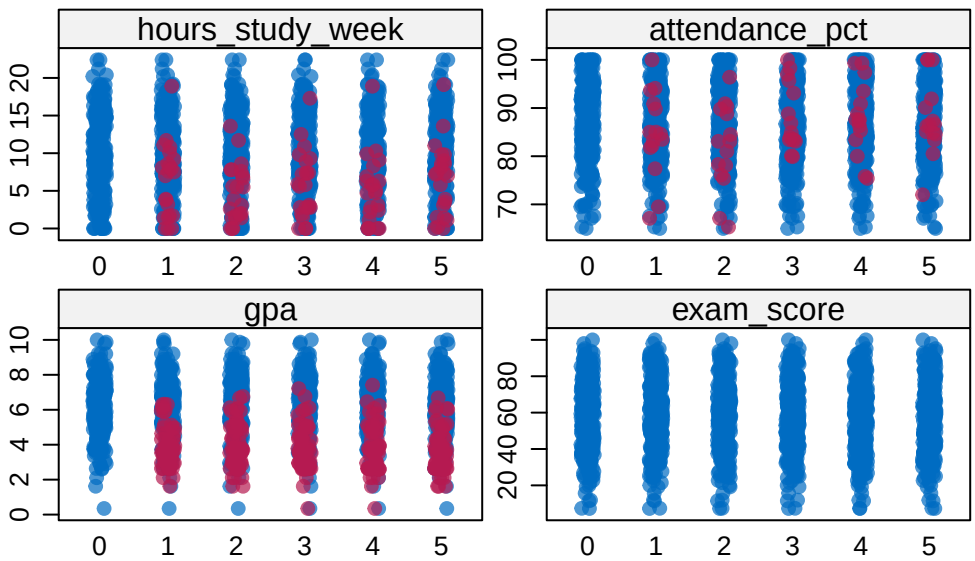
```
attendance_pct                  1                0    1           1        1      1
gpa                             1                1    0           1        1      1
exam_score                      1                1    1           0        1      1
program                         1                1    1           1        0      1
shift                           1                1    1           1        1      0
```

- Visualizar V. Cuantitativas

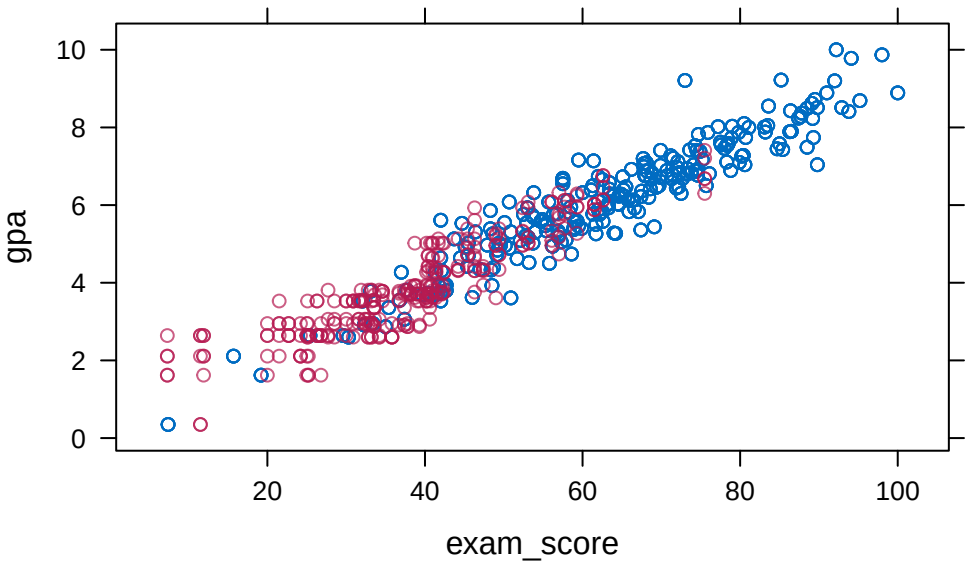
```r
###Visualizar las variables cuantitativas
densityplot(impData)
```



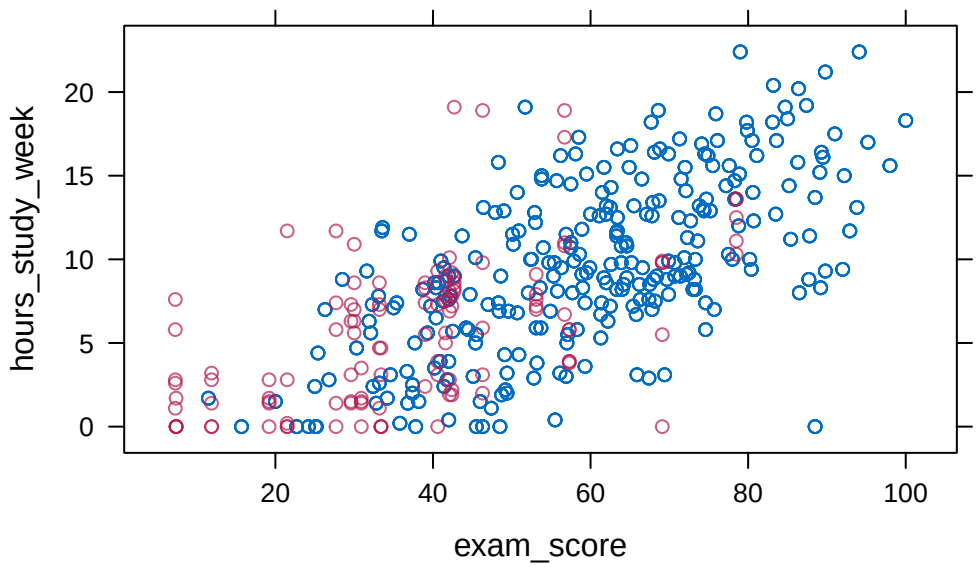

```r
stripplot(impData, pch = 20, cex = 1.2)
```




```
xyplot(impData,gpa~exam_score)
```

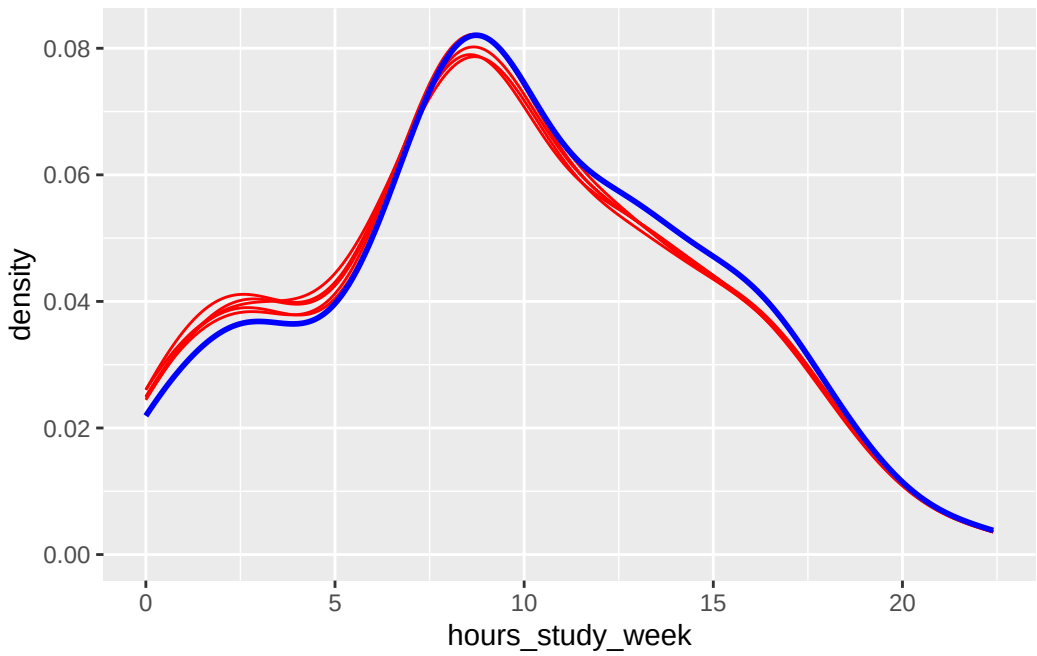



```
xyplot(impData,hours_study_week~exam_score)
```



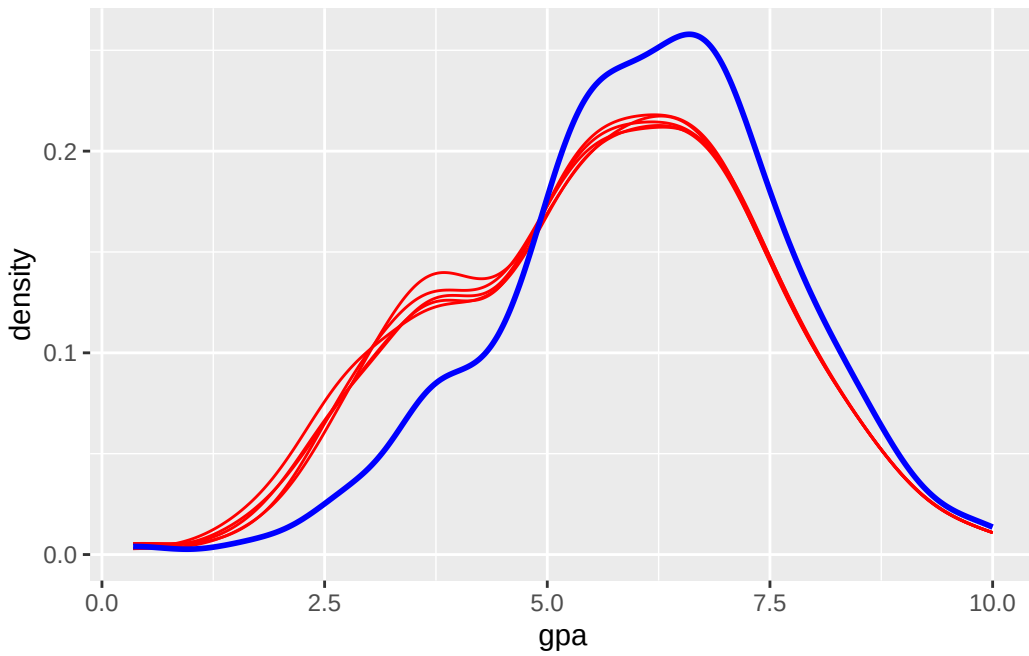

```r
### Visualizar las variables completas
completedData <- complete(impData, "long")

ggplot(completedData, aes(x = hours_study_week)) +
  geom_density(aes(group = .imp), color = "red", alpha = 0.5) +
  geom_density(data = data, aes(x = hours_study_week), color = "blue", linewidth = 1)
```

```
Warning: Removed 23 rows containing non-finite outside the scale range
(`stat_density()`).
```



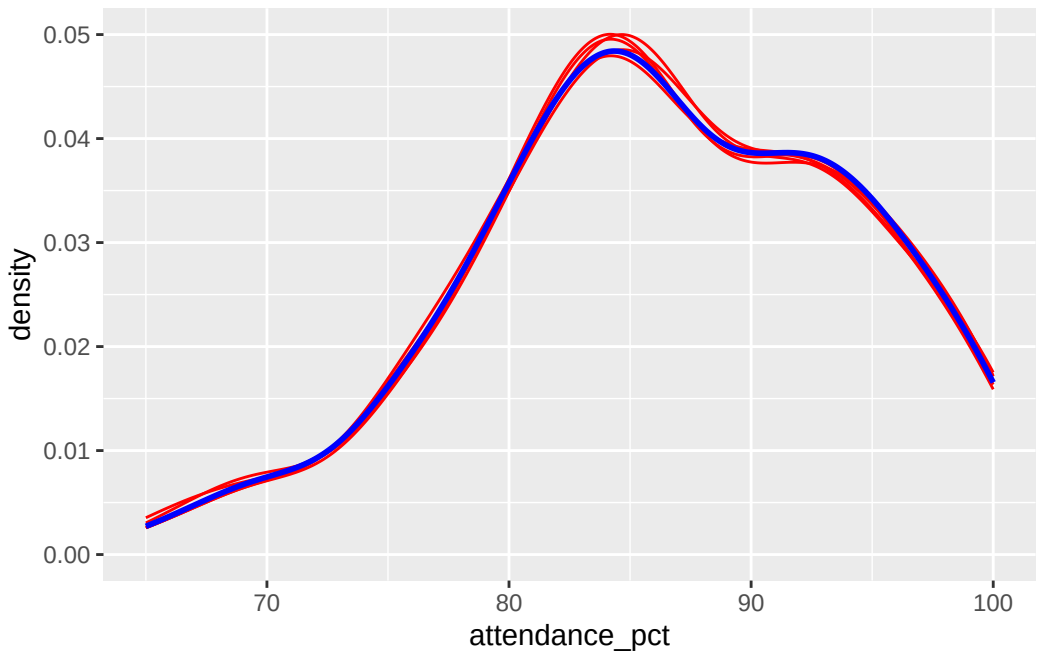

```
ggplot(completedData, aes(x = gpa)) +
  geom_density(aes(group = .imp), color = "red", alpha = 0.5) +
  geom_density(data = data, aes(x = gpa), color = "blue", linewidth = 1)
```

```
Warning: Removed 61 rows containing non-finite outside the scale range
(`stat_density()`).
```



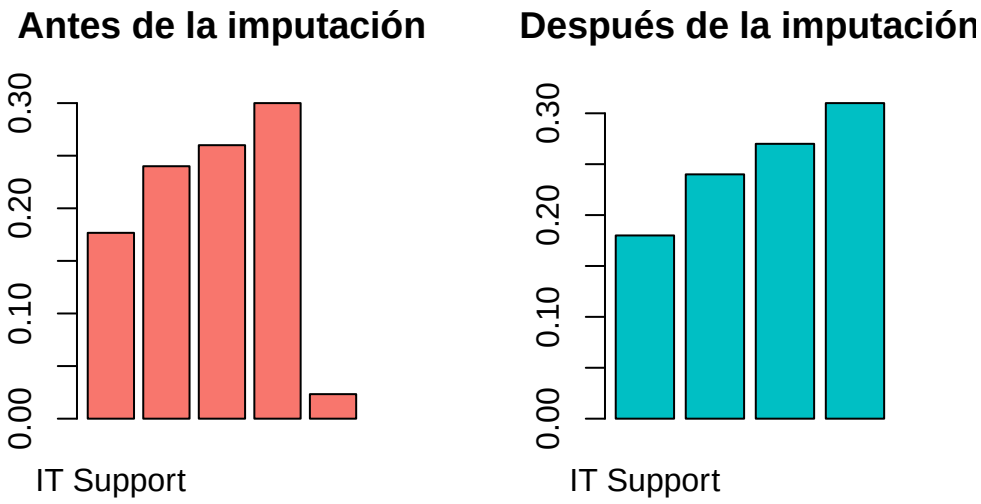

```
ggplot(completedData, aes(x = attendance_pct)) +
  geom_density(aes(group = .imp), color = "red", alpha = 0.5) +
  geom_density(data = data, aes(x = attendance_pct), color = "blue", linewidth = 1)
```

```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_density()`).
```

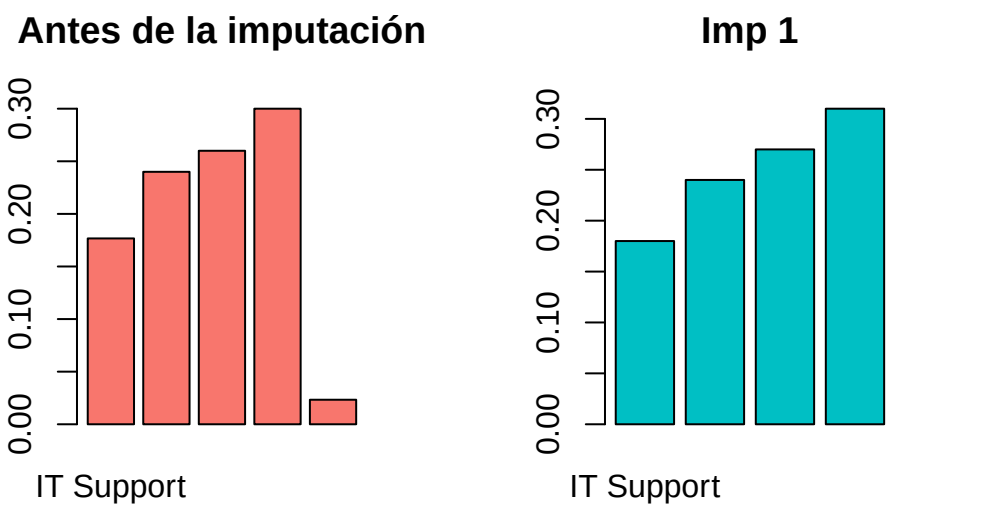


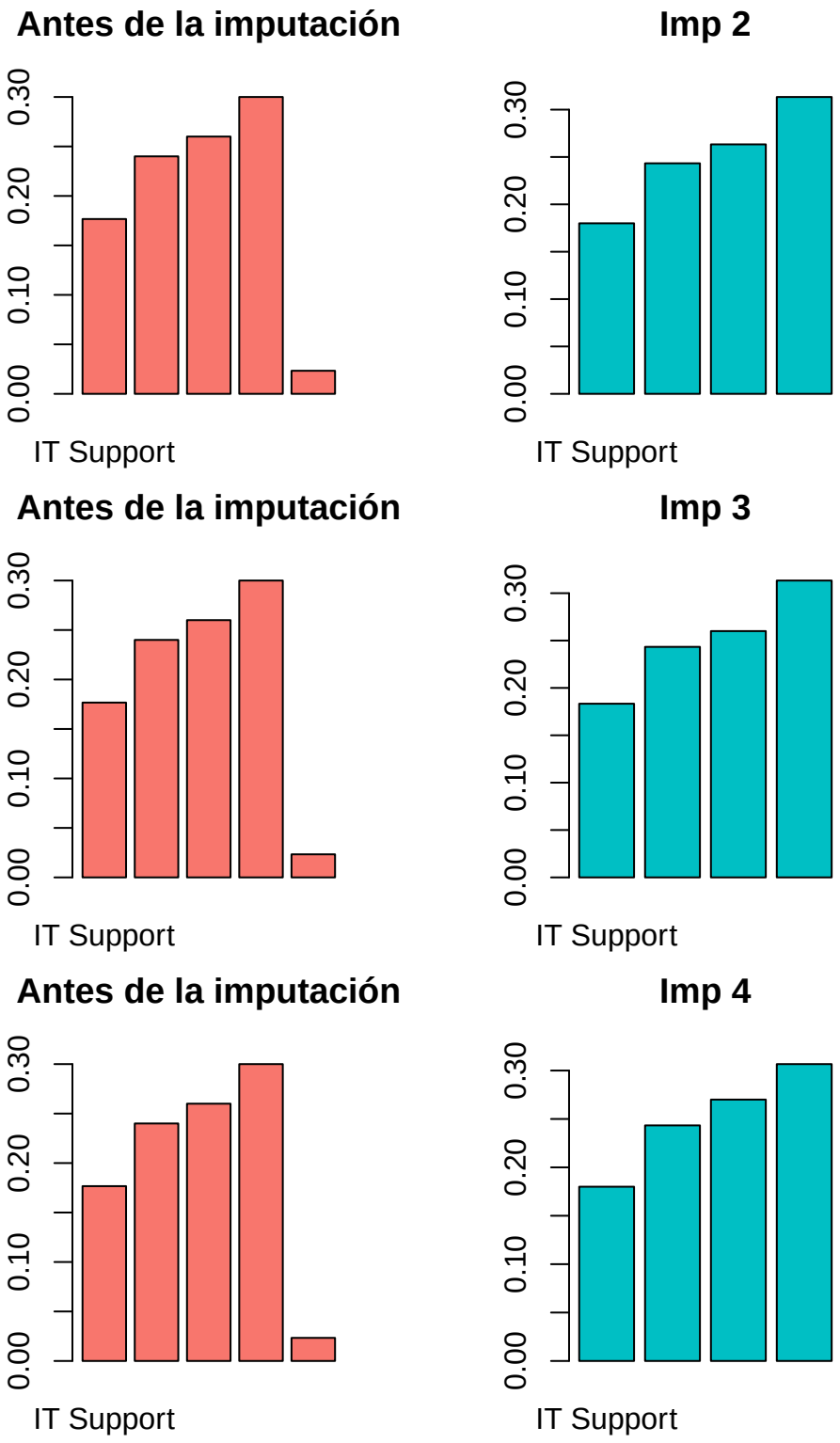
- Visualizar V. Cualitativas

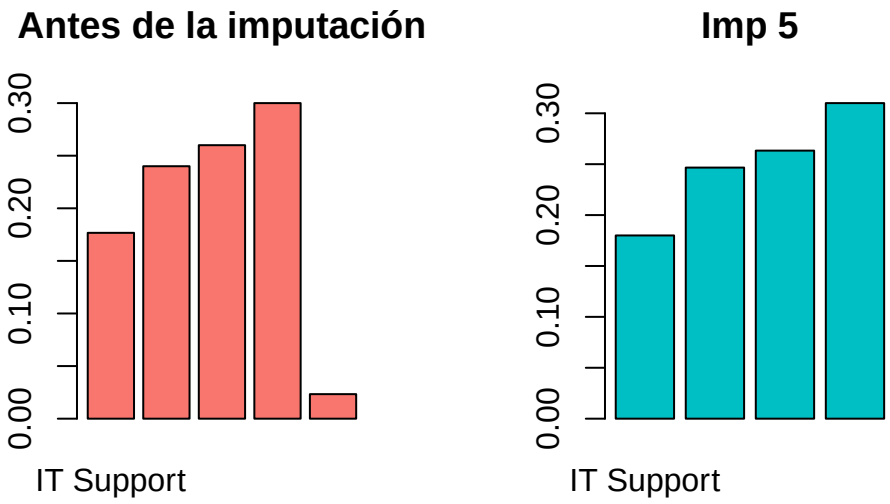
```
###Visualizar las variables cualitativas
completedData <- complete(impData,1)

### Se puede dibujar una a una
par(mfrow = c(1, 2))
barplot(prop.table(table(data$program, useNA = "ifany")),
        main = "Antes de la imputación", col = "#F8766D" )
barplot(prop.table(table(completedData$program)),
        main = "Después de la imputación",col = "#00BFC4")
```




```r
### Así se pueden mirar todas de golpe
for (i in 1:impData$m) {
  par(mfrow = c(1, 2))
  barplot(prop.table(table(data$program, useNA = "ifany")),
          main = "Antes de la imputación", col = "#F8766D")
  comp <- complete(impData, i)
  barplot(prop.table(table(comp$program)),
          main = paste("Imp", i), col = "#00BFC4")
}
```



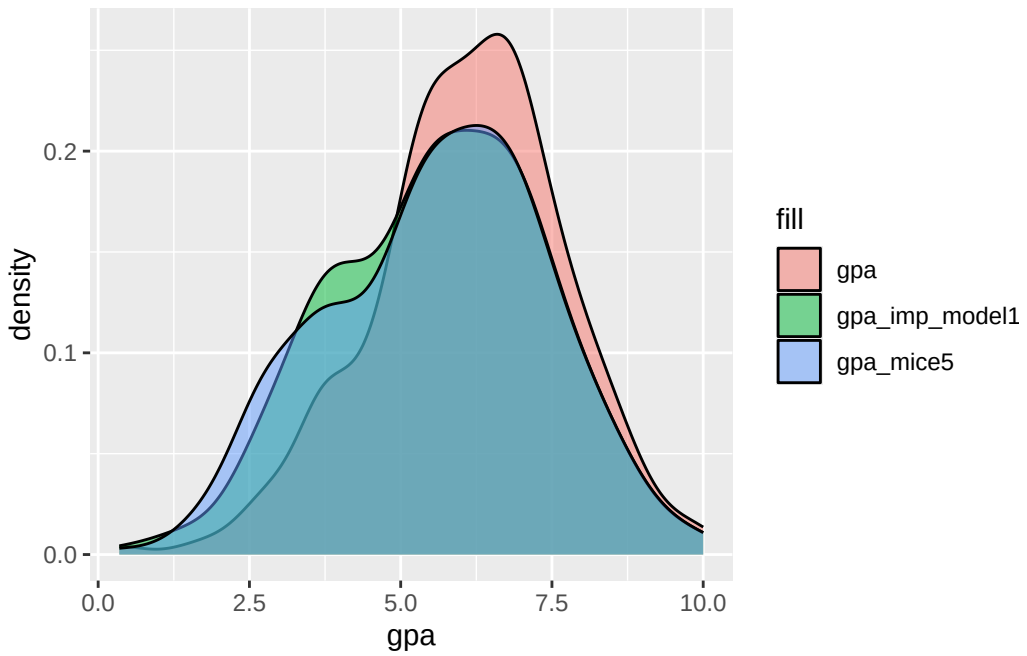

Antes de la imputación
Imp 2
Antes de la imputación
Imp 3
Antes de la imputación
Imp 4
0.00
0.10
0.20
0.30
IT Support



4. Comparación con las otras

```
data$gpa_mice1 <- complete(impData,1)$gpa
data$gpa_mice2 <- complete(impData,2)$gpa
data$gpa_mice3 <- complete(impData,3)$gpa
data$gpa_mice4 <- complete(impData,4)$gpa
data$gpa_mice5 <- complete(impData,5)$gpa


ggplot(data, aes(x = gpa, fill = "gpa")) +
  geom_density(alpha = 0.5) +
  geom_density(aes(x = gpa_imp_model1, fill = "gpa_imp_model1"),alpha = 0.5) +
   geom_density(aes(x = gpa_mice5, fill = "gpa_mice5"),alpha = 0.5)
```

```
Warning: Removed 61 rows containing non-finite outside the scale range
(`stat_density()`).
```


density
0.2
0.1
0.0
0.0
2.5
5.0
7.5
10.0
gpa
fill
gpa
gpa_imp_model1
gpa_mice5